M339G: April 1st, 2026.

Bivariate Normal in Matrix Notation.

Consider a bivariate normal pair (U,V).

In 2D: we place the means into a vector

$$\mu := \begin{pmatrix} \mu_U \\ \mu_V \end{pmatrix}$$ anything in $\mathbb{R}^2$

The variances / covariances are placed into a matrix:

$$\Sigma := \begin{bmatrix} \sigma_U^2 & \sigma_U\sigma_V\rho \\ \sigma_U\sigma_V\rho & \sigma_V^2 \end{bmatrix}$$ (positive definite)

In 1D:

$$f_W(w) = \frac{1}{\sigma_W\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{w-\mu_W}{\sigma_W}\right)^2}$$

Then, the joint density of (U,V) can be written as:

$$\boxed{f_{U,V}(u,v) = \frac{1}{2\pi} \cdot \frac{1}{(\det(\Sigma))^{1/2}} \exp\left(-\frac{1}{2}\underbrace{\begin{pmatrix} u-\mu_U \\ v-\mu_V \end{pmatrix}^T}_{1\times 2}\underbrace{\Sigma^{-1}}_{2\times 2}\underbrace{\begin{pmatrix} u-\mu_U \\ v-\mu_V \end{pmatrix}}_{2\times 1}\right)}$$

Task: What do you get for (X,Z) bivariate standard normal w/ correlation ρ?

Multivariate Normal Density.

Let $\mathbb{X} = (X_1, X_2, \dots, X_p)^T$ be

$$\text{Normal}\left(\text{mean} = \mu = (\mu_1, \dots, \mu_p)^T, \ \Sigma = \begin{bmatrix} \sigma_1^2 & & \text{Cov} \\ & \ddots & \\ \text{Cov} & & \sigma_p^2 \end{bmatrix}\right)$$

w/ Σ positive definite

Then,

$$\boxed{f_X(\underbrace{x_1, \dots, x_p}_{x}) = \frac{1}{(2\pi)^{p/2}} \cdot \frac{1}{(\det(\Sigma))^{1/2}} \exp\left(-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)\right) \quad \text{for all } x \in \mathbb{R}^p}$$